PART-3 MATHEMATICS

SECTION -1 (Only One Option Correct Type)

This section contains 10 multiple choice questions. Each question has four choices (A), B), (C) and (D) out of which only one option is correct.

---

41. The function $y = f(x)$ is the solution of the differential equation

$$\frac{dy}{dx} + \frac{xy}{x^2 - 1} = \frac{x^4 + 2x}{\sqrt{1 - x^2}}$$

in $(-1, 1)$ satisfying $f(0) = 0$. Then

$$\int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} f(x)\ dx$$

is

(A) $\frac{\pi}{3} - \frac{\sqrt{3}}{2}$ (B) $\frac{\pi}{3} - \frac{\sqrt{3}}{4}$ (C) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$ (D) $\frac{\pi}{6} - \frac{\sqrt{3}}{2}$

Answer (B)

42. The following integral

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \operatorname{cosec}\ x)^{17}\ dx$$

is equal to

(A) $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16} du$

(B) $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17} du$

(C) $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17} du$

(D) $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16} du$

Answer (A)

43. Coefficient of x^{11} in the expansion of $(1+x^2)^4\ (1+x^3)^7(1+x^4)^{12}$ is

(A) 1051

(B) 1106

(C) 1113

(D) 1120

Answer (C)

44. Let $f:[0,2] \to \mathbb{R}$ be a function which is continuous on $[0,2]$ and is differentiable on $(0,2)$ with $f(0) = 1$. Let

$$F(x) = \int_0^{x^2} f(\sqrt{t})\, dt$$

for $x \in [0,2]$. If $F'(x) = f'(x)$ for all $x \in (0,2)$, then $F(2)$ equals

(A) $e^2 - 1$

(B) $e^4 - 1$

(C) $e - 1$

(D) e^4

Answer (B)

45. The common tangents to the circle $x^2 + y^2 = 2$ and the parabola $y^2 = 8x$ touch the circle at the points P, Q and the parabola at the points R, S. Then the area of the quadrilateral $PQRS$ is

(A) 3
(B) 6
(C) 9
(D) 15

Answer (D)

46. For $x \in (0,\ \pi)$, the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has

(A) infinitely many solutions

(B) three solutions

(C) one solution

(D) no solution

Answer (D)

47. In a triangle the sum of two sides is x and the product of the same two sides is y. If $x^2 - c^2 = y$, where c is the third side of the triangle, then the ratio of the in-radius to the circum-radius of the triangle is

(A) $\frac{3y}{2x(x+c)}$

(B) $\frac{3y}{2c(x+c)}$

(C) $\frac{3y}{4x(x+c)}$

(D) $\frac{3y}{4c(x+c)}$

Answer (B)

48. Six cards and six envelopes are numbered $1, 2, 3, 4, 5, 6$ and cards are to be placed in envelopes so that each envelope contains exactly one card and no card is placed in the envelope bearing the same number and moreover the card numbered 1 is always placed in envelope numbered 2. Then the number of ways it can be done is

(A) 264

(B) 265

(C) 53

(D) 67

Answer (C)

49. Three boys and two girls stand in a queue. The probability, that the number of boys ahead of every girl is at least one more than the number of girls ahead of her, is

(A) $\frac{1}{2}$ **(B)** $\frac{1}{3}$ **(C)** $\frac{2}{3}$ **(D)** $\frac{3}{4}$

Answer (A)

50. The quadratic equation $p(x) = 0$ with real coefficients has purely imaginary roots. Then the equation

$$p\big(p(x)\big) = 0$$

has

(A) only purely imaginary roots

(B) all real roots

(C) two real and two purely imaginary roots

(D) neither real nor purely imaginary roots

Answer (D)

Section -2 Comprehension Type

This section contains 3 paragraphs each describing theory, experiments, data etc. Six questions relate to the three paragraphs with two questions on each paragraph. Each question has only one correct answer among the four given options (A), (B), (C) and (D).

Paragraph For Questions 51 and 52

Let a, r, s, t be nonzero real numbers. Let $P(at^2, 2at)$, Q, $R(ar^2, 2ar)$ and $S(as^2, 2as)$ be distinct points on the parabola $y^2 = 4ax$. Suppose that PQ is the focal chord and lines QR and PK are parallel, where K is the point $(2a, 0)$.

51. The value of r is

Choices:

(A) $-\frac{1}{t}$

(B) $\frac{t^2+1}{t}$

(C) $\frac{1}{t}$

(D) $\frac{t^2-1}{t}$

Answer (D)

52. If $st = 1$, then the tangent at P and the normal at S to the parabola meet at a point whose ordinate is

(A) $\frac{(t^2+1)^2}{2t^3}$

(B) $\frac{a(t^2+1)^2}{2t^3}$

(C) $\frac{a(t^2+1)^2}{t^3}$

(D) $\frac{a(t^2+2)^2}{t^3}$

Answer (B)

Paragraph For Question 53 and 54

Given that for each $a \in (0,1)$,

$$\lim_{h \to 0^+} \int_{h}^{1-h} t^{-a}(1-t)^{a-1} dt$$

exists. Let this limit be $g(a)$. In addition, it is given that the function $g(a)$ is differentiable on $(0,1)$.

53. The value of $g\left(\frac{1}{2}\right)$ is

(A) π

(B) 2π

(C) $\frac{\pi}{2}$

(D) $\frac{\pi}{4}$

Answer (A)

54. The value of $g'\left(\frac{1}{2}\right)$ is

(A) $\frac{\pi}{2}$

(B) π

(C) $-\frac{\pi}{2}$

(D) 0

Answer (D)

Paragraph For Questions 55 and 56

Box 1 contains three cards bearing numbers $1, 2, 3$; box 2 contains five cards bearing numbers $1, 2, 3, 4, 5$; and box 3 contains seven cards bearing numbers $1, 2, 3, 4, 5, 6, 7$. A card is drawn from each of the boxes. Let x_i be the number on the card drawn from the ith box, $i = 1,2,3$.

55. The probability that $x_1 + x_2 + x_3$ is odd, is

(A) $\frac{29}{105}$

(B) $\frac{53}{105}$

(C) $\frac{57}{105}$

(D) $\frac{1}{2}$

Answer (B)

56. The probability that x_1, x_2, x_3 are in an arithmetic progression, is

(A) $\frac{9}{105}$

(B) $\frac{10}{105}$

(C) $\frac{11}{105}$

(D) $\frac{7}{105}$

Answer (C)

Section – 3 (Matching List Type)

This section contains four questions, each having two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (A), (B), (C) and (D), out of which one is correct.

57. Let $z_k = \cos\left(\frac{2k\pi}{10}\right) + i \sin\left(\frac{2k\pi}{10}\right)$; $k = 1, 2, \ldots, 9$.

List I	List II
P. For each z_k there exists a z_j such that $z_k \cdot z_j$	1. True
Q. There exists a $k \in \{1,2, \ldots, 9\}$ such that $z_1 \cdot z =$	**2. False**
z_k has no solution z in the set of complex numbers.	
R. $\frac{\lvert 1-z_1\rvert\lvert 1-z_2\rvert \cdots \lvert 1-z_9\rvert}{10}$ equals	3. 1
S. $1 - \sum_{k=1}^{9} \cos\left(\frac{2k\pi}{10}\right)$ equals	**4. 2**

P Q R S
(A) 1 2 4 3
(B) 2 1 3 4

(C) 1 2 3 4

(D) 2 1 4 3

Answer (C)

58.

List I	List II
P. The number of polynomials $f(x)$ with non-negative integer coefficients of degree ≤ 2, satisfying $f(0) = 0$ and $\int_0^1 f(x)dx = 1$, is	1. 8
Q. The number of points in the interval $[-\sqrt{13}, \sqrt{13}]$ at which $f(x) = \sin(x^2) + \cos(x^2)$ attains its maximum value, is	2. 2
R. $\int_{-2}^{2} \frac{3x^2}{(1+e^x)}\ dx$ equals	**3. 4**
S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}{\left(\int_{0}^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}$ equals	**4. 0**

P Q R S
(A) 3 2 4 1
(B) 2 3 4 1

(C) 3 2 1 4

(D) 2 3 1 4

Answer (D)

59.

List I	List II
P. Let $y(x) = \cos(3\ \cos^{-1} x)\,, x \in [-1,1], x \neq \pm\frac{\sqrt{3}}{2}$. Then $\frac{1}{y(x)}\left\{(x^2 - 1)\ \frac{d^2y(x)}{dx^2} + x\ \frac{dy(x)}{dx}\right\}$ equals	1. 1
Q. Let $A_1, A_2, \ldots, A_n$ $(n > 2)$ be the vertices of a regular polygon of n sides with its centre at the origin. Let $\overrightarrow{a_k}$ be the position vector of the point $A_k,\ k = 1,2, \ldots, n$. If $\lvert\sum_{k=1}^{n-1}(\overrightarrow{a_k} \times \overrightarrow{a_{k+1}})\rvert = \lvert\sum_{k=1}^{n-1}(\overrightarrow{a_k} \cdot \overrightarrow{a_{k+1}})\rvert$, then the minimum value of n is	2. 2
R. If the normal from the point $P(h, 1)$ on the ellipse $\frac{x^2}{6} + \frac{y^2}{3} = 1$ is perpendicular to the line $x + y = 8$, then the value of h is	**3. 8**
S. Number of positive solutions satisfying the equation $\tan^{-1}\left(\frac{1}{2x+1}\right) + \tan^{-1}\left(\frac{1}{4x+1}\right) = \tan^{-1}\left(\frac{2}{x^2}\right)$ is	**4. 9**

	P	Q	R	S
(A)	4	3	2	1
(B)	2	4	3	1
(C)	4	3	1	2
(D)	2	4	1	3

Answer (A)

60. Let $f_1: \mathbb{R} \to \mathbb{R}$, $f_2: [0, \infty) \to \mathbb{R}$, $f_3: \mathbb{R} \to \mathbb{R}$ and $f_4: \mathbb{R} \to [0, \infty)$ be defined by

$$f_1(x) = \begin{cases} |x| & \text{if } x < 0, \\ e^x & \text{if } x \geq 0; \end{cases}$$

$$f_2(x) = x^2;$$

$$f_3(x) = \begin{cases} \sin x & \text{if } x < 0, \\ x & \text{if } x \geq 0 \end{cases}$$

and

$$f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if } x < 0, \\ f_2(f_1(x)) - 1 & \text{if } x \geq 0. \end{cases}$$

List II

P. f_4 is

Q. f_3 is

R. $f_2 o f_1$ is

S. f_2 is

List I

1. onto but not one-one

2. neither continuous nor one-one

3. differentiable but not one-one

4. continuous and one-one

	P	Q	R	S
(A)	3	1	4	2
(B)	1	3	4	2
(C)	3	1	2	4
(D)	1	3	2	4

Answer (D)